What is cybersecurity? Explain the CIA triad with one real-world example for each part.

Cybersecurity is the practice of protecting computers, networks, applications, and digital information from unauthorized access, cyberattacks, theft, and damage.

Its main purpose is to keep information secure, maintain system reliability, and protect users from cyber threats.

Main objectives of cybersecurity

- Protect sensitive and confidential information.
- Prevent unauthorized access to systems and networks.
- Detect and respond to cyberattacks.
- Maintain the accuracy and availability of data.
- Reduce security risks and protect user privacy.

Common types of cybersecurity

Type	Description	Example
Network security	Protects networks and traffic.	Firewalls
Application security	Protects software from vulnerabilities.	Secure coding
Information security	Protects sensitive information.	Data encryption
Endpoint security	Protects computers and devices.	Antivirus software
Cloud security	Protects cloud systems and data.	Cloud access controls

CIA Triad

The CIA triad is a fundamental cybersecurity model consisting of three principles: **Confidentiality, Integrity, and Availability**. It helps organizations protect information and systems against security threats.

A. Confidentiality

Definition: Confidentiality ensures that information is accessible only to authorized users.

Real-world example: A hospital allows only authorized doctors and healthcare staff to access patients' medical records. If an unauthorized person views these records, confidentiality is compromised.

Security measures:

- Strong passwords and multi-factor authentication.
- Encryption of sensitive information.
- Access controls and user permissions.

B. Integrity

Definition: Integrity ensures that information remains accurate, complete, and trustworthy, and that unauthorized changes are prevented or detected.

Real-world example: A bank maintains accurate customer account balances. If an attacker changes a balance without authorization, the integrity of the financial records is compromised.

Security measures:

- File hashing and integrity monitoring.
- Audit logs and access controls.
- Digital signatures and change management.

C. Availability

Definition: Availability ensures that authorized users can access systems, applications, and information when needed.

Real-world example: An online banking service becomes unavailable because of a server failure or a denial-of-service attack. Customers cannot access their accounts, resulting in an availability failure.

Security measures:

- Regular backups.
- Redundant systems and servers.
- System monitoring and disaster recovery.
- Denial-of-service protection.

CIA Triad Comparison Table

Principle	Main purpose	Example of failure	Protection
Confidentiality	Prevent unauthorized disclosure.	A hacker steals customer information.	Encryption and access control.
Integrity	Prevent or detect unauthorized changes.	A bank record is modified illegally.	Hashing and audit logs.
Availability	Ensure timely access to systems.	A website becomes unavailable.	Backups and redundancy.

Importance of CIA Triad

The CIA triad helps organizations identify security requirements and select appropriate protective measures.

For example, an online shopping website must protect customers' personal information (confidentiality), ensure order and payment records remain accurate (integrity), and keep its services accessible to customers (availability).

All three principles are important because protecting information from disclosure alone does not guarantee that it remains accurate or accessible.

Conclusion

Cybersecurity protects digital systems and information from unauthorized access, modification, and disruption. The CIA triad provides three essential security objectives:

- **Confidentiality:** Keep information private.
- **Integrity:** Keep information accurate and trustworthy.
- **Availability:** Keep systems and information accessible when required.

Understanding these principles provides a foundation for learning Linux security, networking, SOC analysis, and vulnerability assessment and penetration testing (VAPT).